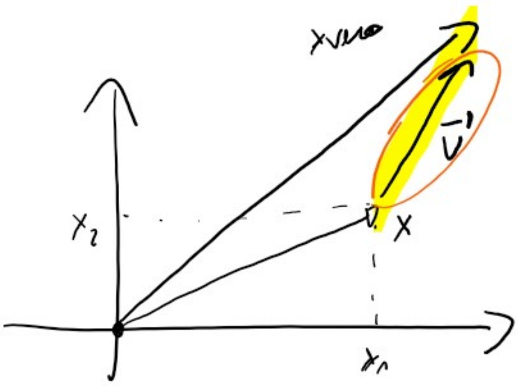
26/3/2025

mercoledì 26 marzo 2025 11:11

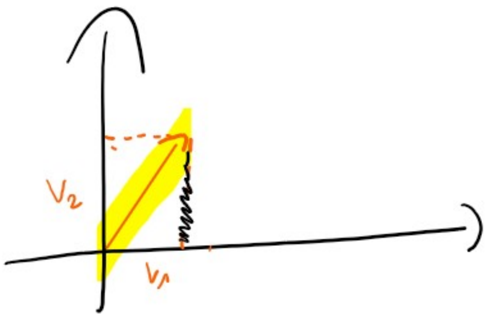
$$r = |x - xvera|$$



$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad xvera = \begin{pmatrix} xvera_1 \\ xvera_2 \end{pmatrix}$$

$$\vec{v} = \vec{x} - \vec{x}_{vera} = \begin{pmatrix} \overbrace{x_1 - xvera_1}^{v_1} \\ \underbrace{x_2 - xvera_2}_{v_2} \end{pmatrix}$$

$$\vec{x}_{vera} = \vec{x} + \vec{v}$$



$$\|\vec{v}\|_\infty = \max\{|v_1|, |v_2|\} = |v_2|$$

$$\|\vec{v}\|_2 = \sqrt{v_1^2 + v_2^2}$$

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gaussNaive.m

```
function [A,b,deter] = gaussNaive(A,b)
deter=1; n=size(A,1); C=norm(A);
for k=1:n-1
    if A(k,k)==0
        deter=[]; disp('Metodo non applicabile');
        return
    elseif ( abs(A(k,k) )<eps*C )
        disp('ATTENZIONE!!! POSSIBILE ELEMENTO NULLO SULLA DIAGONALE!!!');
    end
    deter=deter*A(k,k);
    for i=k+1:n
        m=A(i,k)/A(k,k);
        A(i,k)=0;
        A(i,k+1:n)=A(i,k+1:n)-m*A(k,k+1:n); %for j=k+1:n
        b(i)=b(i)-m*b(k);
    end
end
```

```
deter=deter*A(n,n);
if A(n,n)==0
    deter=[]; disp('Attenzione det(A)=0')
    return
elseif ( abs(A(n,n) )<eps*C )
    disp('ATTENZIONE!!! POSSIBILE MATRICE SINGOLARE!!!');
end
```

e